

NGIT/KMEC Project School

II - I Project School -NGIT Project List (2026-27 ODD Sem)

	Title	ShortTitle	Category	TechStack	Reference Link
1	Detection of Dyslexia Through Images of Handwriting using Hybrid AI Approach	Dyslexia using CNN-SVM	NEURO	Python, TensorFlow/Keras, CNN, SVM, Random Forest, OpenCV, NumPy, Pandas, Scikit-Learn	https://drive.google.com/drive/u/0/folders/1XwFO6fJJbiqSziEO2TUZVt-LUr05YYfD
2	Predicting risk of dyslexia with an online gamified test	Dyslexia using OGT	NEURO	Python, Pandas, NumPy, Scikit-Learn, Random Forest, Matplotlib,	https://drive.google.com/drive/u/0/folders/1XwFO6fJJbiqSziEO2TUZVt-LUr05YYfD
3	Detection of developmental dyslexia with machine learning using eye movement data	Dyslexia using ML	NEURO	Python, Pandas, NumPy, Scikit-Learn, Random Forest, Support Vector Machine (SVM), Eye Tracking Data Analysis	https://drive.google.com/drive/u/0/folders/1XwFO6fJJbiqSziEO2TUZVt-LUr05YYfD
4	Machine Learning the Voltage of Electrode Materials in Metal-ion Batteries	ML-MiB	CRSYTAL	Python, Pandas, NumPy, Scikit-Learn, PyTorch, Random Forest, XGBoost,	https://drive.google.com/drive/u/0/folders/1XwFO6fJJbiqSziEO2TUZVt-LUr05YYfD
5	Machine Learning-Assisted Discovery of Solid Li-Ion Conducting Materials	ML-SLICM	CRSYTAL	Python, Pandas, NumPy, Scikit-Learn, PyTorch, Logistic Regression, Random Forest (Enhancement), XGBoost (Enhancement)	https://drive.google.com/drive/u/0/folders/1XwFO6fJJbiqSziEO2TUZVt-LUr05YYfD

6	RanSMAP: Open dataset of Ransomware Storage and Memory Access Patterns for creating deep learning based ransomware detectors.	RanSMAP	Cyber	"Python Google Colab RanSMAP Dataset Pandas NumPy TensorFlow / Keras Scikit-learn Matplotlib Seaborn "	https://drive.google.com/drive/u/0/folders/1IG7Bji-lbzdPZFgd4SUEwsPaxx1lrJ9W
7	Cyber Threat Detection Based on Artificial Neural Networks Using Event Profiles	CyberThreat Detection Using AN	Cyber	Python, TensorFlow, Big Data Platform, AI-SIEM, IDS/IPS event logs, SIEM logs	https://drive.google.com/drive/u/0/folders/1IG7Bji-lbzdPZFgd4SUEwsPaxx1lrJ9W
8	Retrieval Augmented Generation for Robust Cyber Defense	RAG-Cyber	Cyber	Python, Neo4j, Cypher Query Language, Knowledge Graphs, RAG, LLM	https://drive.google.com/drive/u/0/folders/1IG7Bji-lbzdPZFgd4SUEwsPaxx1lrJ9W
9	Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot	Enhancing SO-RAG	Cyber	"Python Google Colab RanSMAP Dataset Pandas NumPy Scikit-learn Matplotlib Seaborn "	https://drive.google.com/drive/u/0/folders/1IG7Bji-lbzdPZFgd4SUEwsPaxx1lrJ9W
10	Retrieval-Augmented Semantic Mapping for Vulnerability Detection via Multi-View Code Similarity	Vulnerability Detection-RAG	Cyber	"Python Google Colab RanSMAP Dataset Pandas NumPy TensorFlow / Keras Scikit-learn Matplotlib Seaborn "	https://drive.google.com/drive/u/0/folders/1IG7Bji-lbzdPZFgd4SUEwsPaxx1lrJ9W

11	AI-Based Prediction of Lithium-Ion Battery Cathode Performance Using Explainable Machine Learning	ML_Voltage	Crystal	Python, Pandas, NumPy, Regular Expressions, Matplotlib, Seaborn, Plotly, Scikit-Learn, XGBoost, Logistic Regression	https://drive.google.com/drive/u/0/folders/119FKv4DoXJ5NpHnBUGlFgH2EVMCz4gBb
12	GlassFormAI: Explainable Machine Learning-Based Prediction and Recommendation of Metallic Glass-Forming Alloys	ML_Met_Glass	Crystal	Python, Pandas, NumPy, Regular Expressions, Matplotlib, Seaborn, Plotly, Scikit-Learn, XGBoost, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, XGBoost Classifier, Feature Importance	https://drive.google.com/drive/u/0/folders/119FKv4DoXJ5NpHnBUGlFgH2EVMCz4gBb
13	Multi-modal analysis of infant cry types characterization: Acoustics, body language and brain signals	Infant_Cry	Acoustic	PyTorch, TorchAudio, Librosa, NumPy, SciPy, Praat-Parselmouth, Brainstorm Toolbox, MATLAB, SPSS, scikit-learn, SpeechBrain, PyTorch-Lightning, Transformer, Conformer	https://drive.google.com/drive/u/0/folders/119FKv4DoXJ5NpHnBUGlFgH2EVMCz4gBb
14	A deep learning framework for environmental sound classification by fusing linear and nonlinear features	Sound_Recg_Fusion	Acoustic	Python, PyTorch, TorchAudio, Librosa, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn, Transformer, Conformer	https://drive.google.com/drive/u/0/folders/119FKv4DoXJ5NpHnBUGlFgH2EVMCz4gBb

15	SANADA: A Reproducible Synthetic Acoustic Dataset for Intractable Mixtures, Evaluated with Baseline Models in Separation, Counting, and Tagging	Audio Source Count Estimator	Acoustic	CNNs / Transformers, STFT, Source Counting (regression or classification), PyTorch; Dataset: SANADA (IEEE DataPort)	https://drive.google.com/drive/u/0/folders/1I9FKv4DoXJ5NpHnBUglFgH2EVMCz4gBb
16	Machine-learning-based Structural Analysis of Interactions between Antibodies and Antigens	Antibodies_Antigens_ML	Drug Discovery	"Python, NumPy BioPython scikit-learn TensorFlow / Keras Matplotlib Seaborn, CNN "	https://drive.google.com/drive/u/0/folders/1HGk7QweQjNwIzsqd7CiyC2qKhLRloUFn
17	Improving Synergistic Drug Combination Prediction with Signature-Based Gene Expression Features in Oncology	Synergistic_Drug_Combination_M	Drug Discovery	" Python Google Colab / Jupyter Notebook pandas NumPy scikit-learn XGBoost TensorFlow	https://drive.google.com/drive/u/0/folders/1HGk7QweQjNwIzsqd7CiyC2qKhLRloUFn
18	BiSpec Pairwise AI: guiding the selection of bispecific antibody target combinations with pairwise learning and GPT augmentation	BiSpec_Pairwise	Drug	Python Google Colab / Jupyter Notebook pandas NumPy Scanpy scikit-learn XGBoost Matplotlib Seaborn OpenAI/GPT API or local LLM Streamlit for dashboard	https://drive.google.com/drive/u/0/folders/1HGk7QweQjNwIzsqd7CiyC2qKhLRloUFn

19	Bioactive Molecule Prediction Using Extreme Gradient Boosting	Bioactive_molecule_ML	Drug Discovery	"Python Google Colab pandas NumPy RDKit scikit-learn XGBoost Matplotlib Seaborn Streamlit ChEMBL / DUD-E / COX2 dataset"	https://drive.google.com/drive/u/0/folders/1HGk7QweQjNwIzsqd7CiyC2qKhLRloUFn
20	DietAI24 as a framework for comprehensive nutrition estimation using multimodal large language models	Indian Food Calorie Estimation	NUTRA	Multimodal Large Language Models (LLaVA/BLIP-2/GPT-4V), Vision Transformers (ViT/CLIP), NLP Transformers (BERT), USDA/IFCT Nutrition Databases, Hugging Face, Streamlit/Flask.	https://drive.google.com/drive/u/0/folders/1eQueXsa7vyHXXhnDxWs6pboYQtRY2D2l
21	Natural Language Processing to Extract Information from Portuguese-Language Medical Records	NutriCheck	NUTRA	Hugging Face BioBERT/BioGPT + PyTorch/TensorFlow for NLP, Pandas/NumPy for processing, FastAPI	https://drive.google.com/drive/u/0/folders/1eQueXsa7vyHXXhnDxWs6pboYQtRY2D2l
22	NutriGen: Personalized Meal Plan Generator Leveraging Large Language Models to Enhance Dietary and Nutritional Adherence	NutriGen	NUTRA	Gen AI, Mern Stack, Python	https://drive.google.com/drive/u/0/folders/1eQueXsa7vyHXXhnDxWs6pboYQtRY2D2l
23	Exploring the potential of deep learning models integrating transformer and LSTM in predicting blood glucose levels for T1D patients	NutraT1D	NUTRA	Python 3.10, PyTorch 2.0.0 + CUDA 11.7, Transformer Encoder, LSTM, Bidirectional LSTM (BiLSTM)	https://drive.google.com/drive/u/0/folders/1eQueXsa7vyHXXhnDxWs6pboYQtRY2D2l

24	Development and Validation of an Explainable Machine Learning-Based Prediction Model for Drug-Food Interactions from Chemical Structures	Food_DrugAI	Nutra	Hugging Face BioBERT/BioGPT + PyTorch/TensorFlow for NLP, Pandas/NumPy for processing, FastAPI	https://drive.google.com/drive/u/0/folders/1eQueXsa7vyHXXhnDxWs6pboYQtRY2D2l
----	--	-------------	-------	--	---